PREDICTING YOUTH ALCOHOL USE

A Machine Learning Approach Using Decision Trees & Ensemble Methods

PRESENTED BY: DUONG LE

PROJECT SCOPE

BINARY	DETERMINES WHETHER A YOUTH HAS EVER USED ALCOHOL
MULTI-CLASS	CAPTURES THE FREQUENCY OF USE (NON-USER, LIGHT, MODERATE AND HEAVY)
REGRESSION	AGE OF FIRST USE

ABOUT THE DATA

NATIONAL SURVEY ON DRUG USE AND HEALTH (NSDUH), ADMINISTERED BY SAMHSA

GENERAL CIVILIAN POPULATION AGED 12 AND OLDER IN THE UNITED STATES

LIFETIME USE

PAST-YEAR USE

PAST-MONTH USE

AGE OF FIRST USE

USE DISORDERS

PERCEIVED RISKS

PROTECTIVE FACTORS

DECISION TREES

Splits data recursively based on feature thresholds to arrive at a prediction

Best used when interpretability is the priority

Main weakness: prone to overfitting without pruning or depth constraints

KEY PARAMETERS: DEPTH, SPLITS, LEAF, CRITERION

METRICS:ACCURACY, F-1 SCORE (CLASSIFICATION), MSE/RMSE (REGRESSION)

RANDOM FOREST

Builds many independent trees and aggregates predictions; majority vote (classification) or average (regression)

Best used as a strong generalizable baseline, especially with many features

Reduces overfitting compared to a single tree through aggregation and feature randomness

KEY PARAMETERS: ESTIMATORS, MAXIMUM FEATURES, DEPTH AND WEIGHTS

METRICS:ACCURACY, F-1 SCORE (CLASSIFICATION), MSE/RMSE (REGRESSION)

BAGGING

Trains multiple base models on random data subsets, aggregates their predictions

Best used when the base model has high variance and needs stabilizing

KEY PARAMETERS: MAXIMUM SAMPLES & FEATURES

METRICS: ACCURACY, MACRO F-1, WEIGHTED F-1

BOOSTING

Builds trees sequentially, each correcting the errors of the last

Best used when maximum predictive accuracy is the goal

Key difference from bagging: boosting reduces bias sequentially, bagging reduces variance in parallel

KEY PARAMETERS: ESTIMATORS, LEARNING RATE, DEPTH

METRICS: MSE, RMSE

PREPROCESSING

VARIABLE SELECTION

REDUNDANCY REMOVAL

BINARY ENCODING

DROPPING MISSING DATA

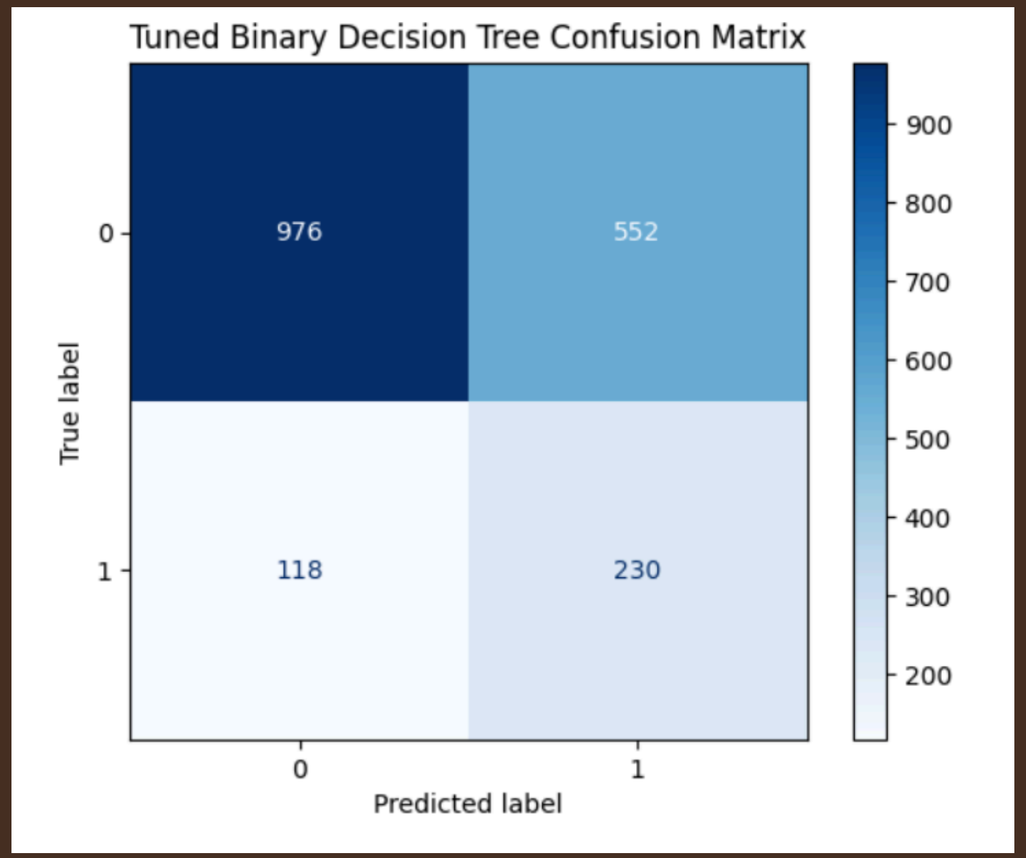
FULL DATASET (~9K ROWS): USED FOR BINARY CLASSIFICATION

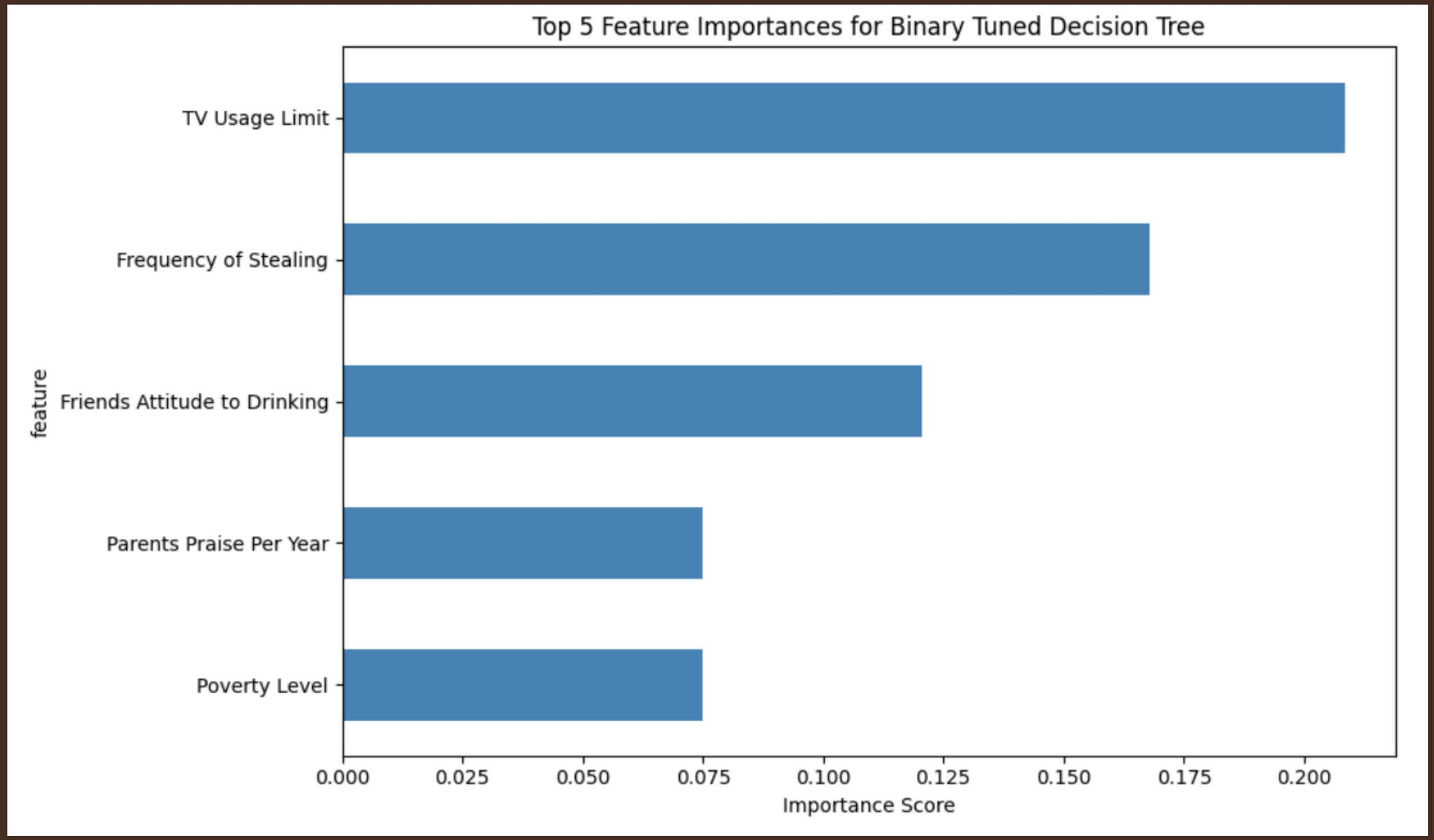
DRINKERS-ONLY DATASET (~2K ROWS): USED FOR MULTICLASS AND REGRESSION

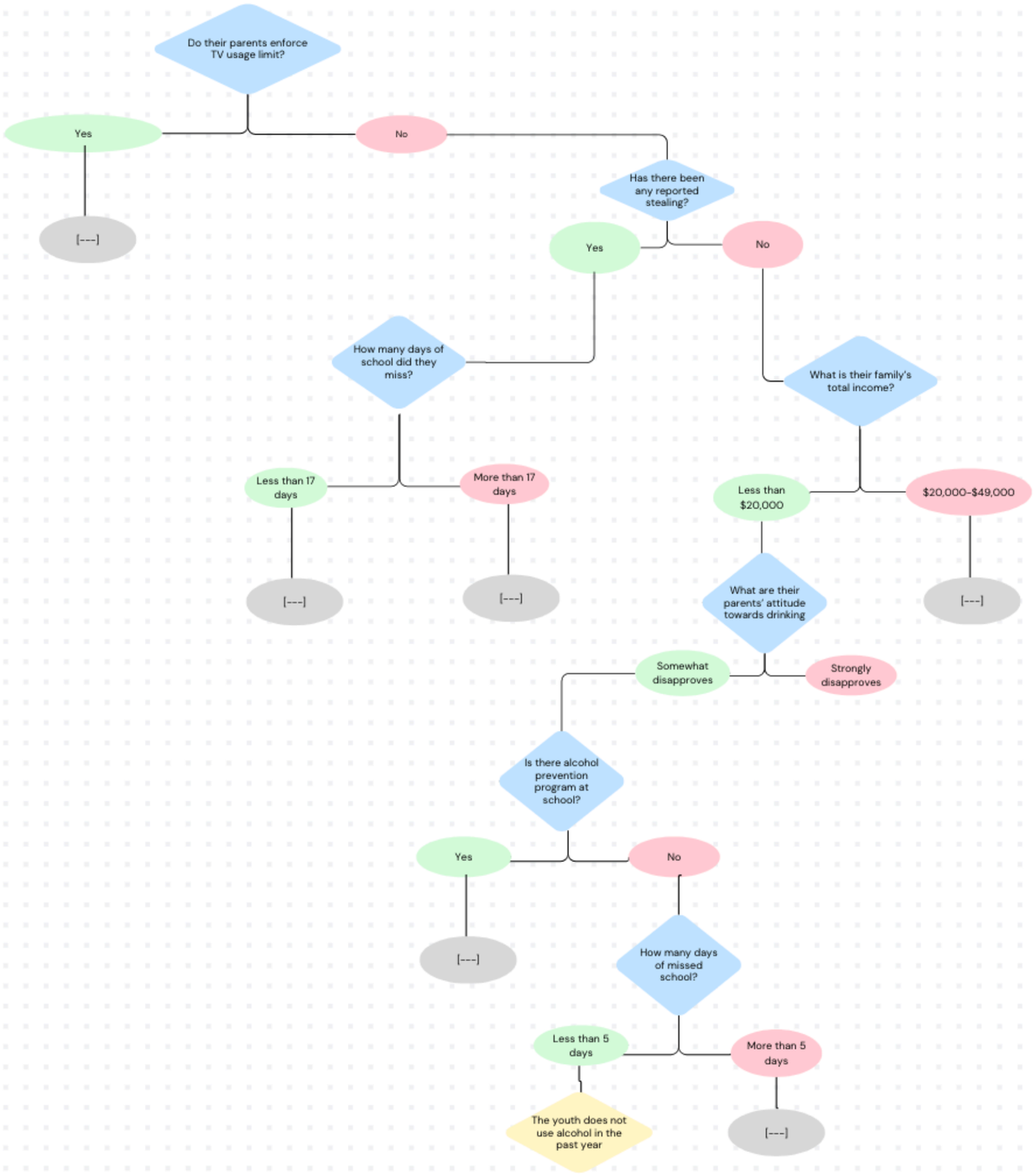
TUNED BINARY DECISION TREE

- Best parameters: max_depth=7, gini criterion, balanced class weights, min_samples_split=20, min_samples_leaf=1
- Train/test difference: 21.65% to 2.31% → overfitting effectively resolved
- Model catches 2 in 3 users (user recall 0.66)
- Overall accuracy reduced to 64.29%
- High false positives (user precision 0.29)

```
Train Accuracy: 0.6660
Test Accuracy: 0.6429
Difference: 0.0231
              precision    recall  f1-score   support

      No Use       0.89      0.64      0.74      1528
         Use       0.29      0.66      0.41       348

    accuracy                           0.64      1876
   macro avg       0.59      0.65      0.58      1876
weighted avg       0.78      0.64      0.68      1876
```




Top 5 Feature Importances for Binary Tuned Decision Tree
TV Usage Limit
Frequency of Stealing
Friends Attitude to Drinking
Parents Praise Per Year
Poverty Level
feature
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Importance Score


Do their parents enforce TV usage limit?
Yes
No
[---]
Has there been any reported stealing?
Yes
No
How many days of school did they miss?
What is their family's total income?
Less than 17 days
More than 17 days
Less than $20,000
$20,000-$49,000
[---]
[---]
What are their parents' attitude towards drinking
[---]
Somewhat disapproves
Strongly disapproves
Is there alcohol prevention program at school?
Yes
No
[---]
How many days of missed school?
Less than 5 days
More than 5 days
The youth does not use alcohol in the past year
[---]

TUNED BINARY RANDOM FOREST

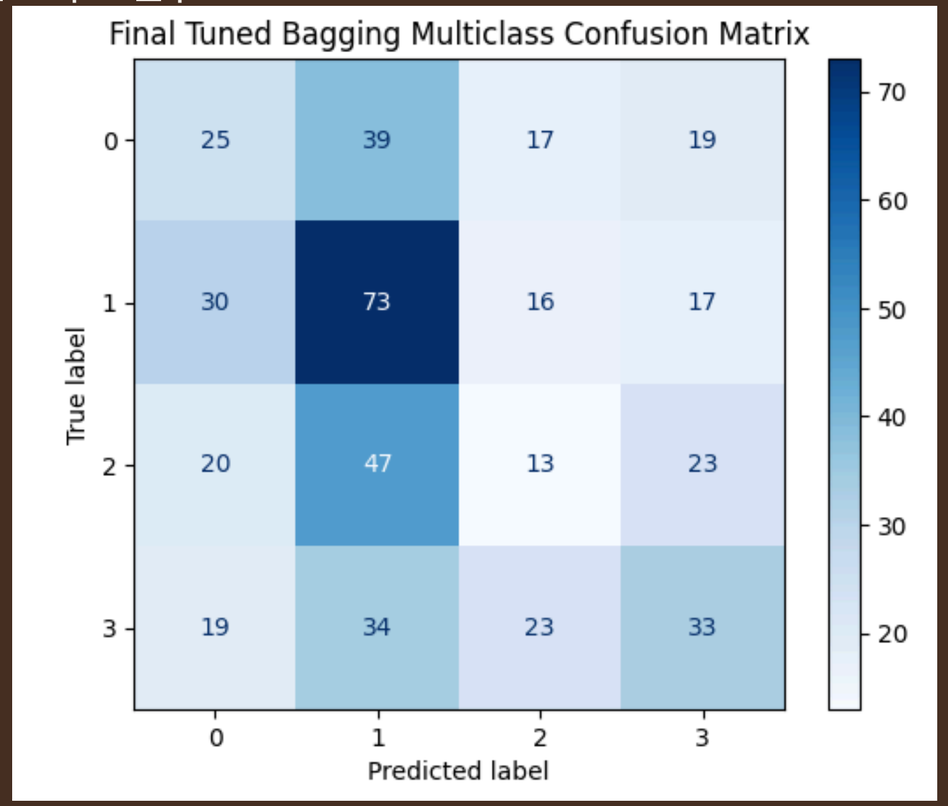
- Best parameters: gini criterion, n_estimators=100, min_samples_leaf=5, min_samples_split=2, no max_depth.
- Class imbalance approach: downsampling (3:1 majority to minority ratio) + upweighting
- Train/test difference: 8.80% → moderate overfitting and higher than tuned decision tree, reflects ensemble complexity
- Model identifies more than half of users (user recall 0.57)
- False positives remain high (user precision 0.31)
- Strong CV performance of all 3 binary models (0.631)
- Lower overall accuracy reflects genuine class imbalance difficulty

```
Train Accuracy: 0.7703
Test Accuracy: 0.6823
Difference:0.0880
              precision    recall  f1-score   support

      No Use       0.88      0.71      0.78      1528
         Use       0.31      0.57      0.40       348

    accuracy                           0.68      1876
   macro avg       0.59      0.64      0.59      1876
weighted avg       0.77      0.68      0.71      1876
```

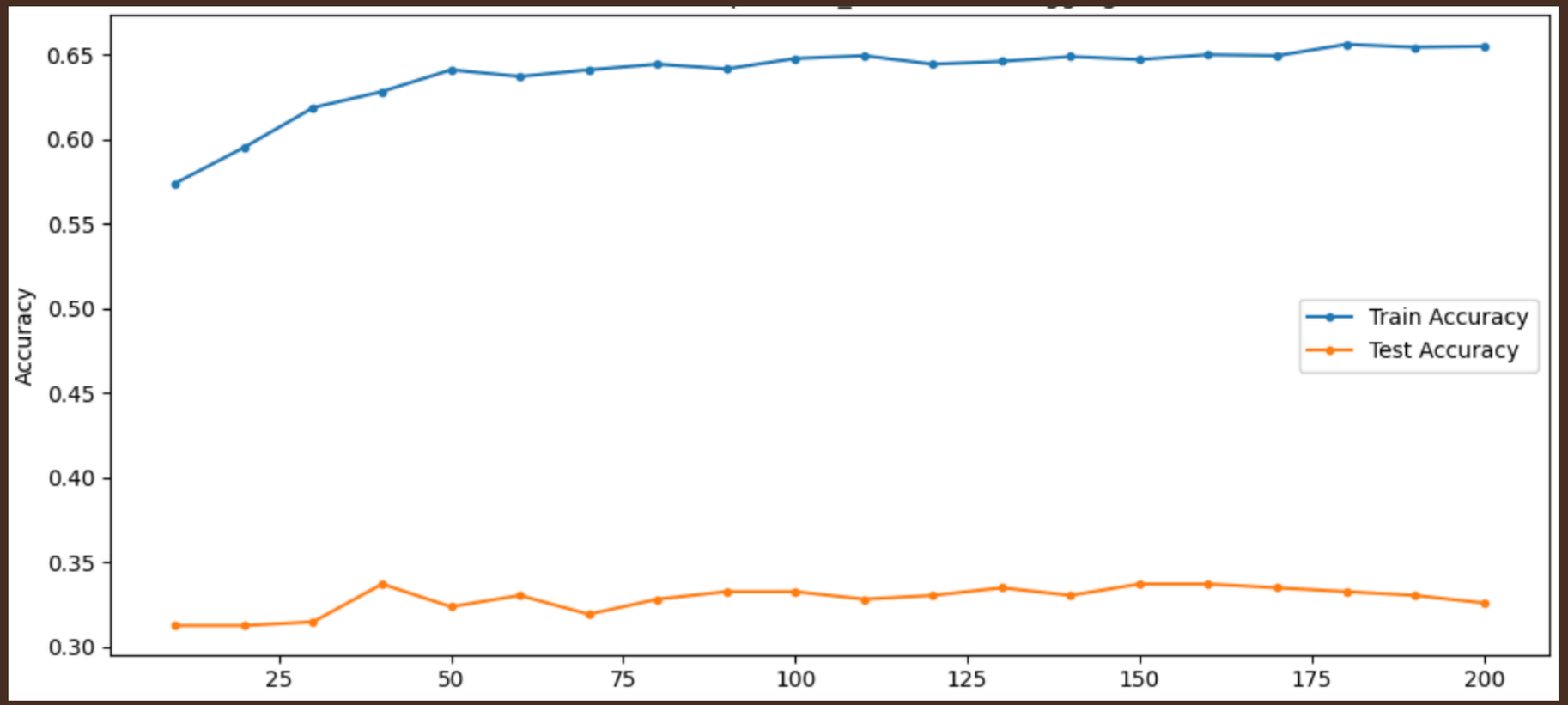
TUNED MULTICLASS BAGGING

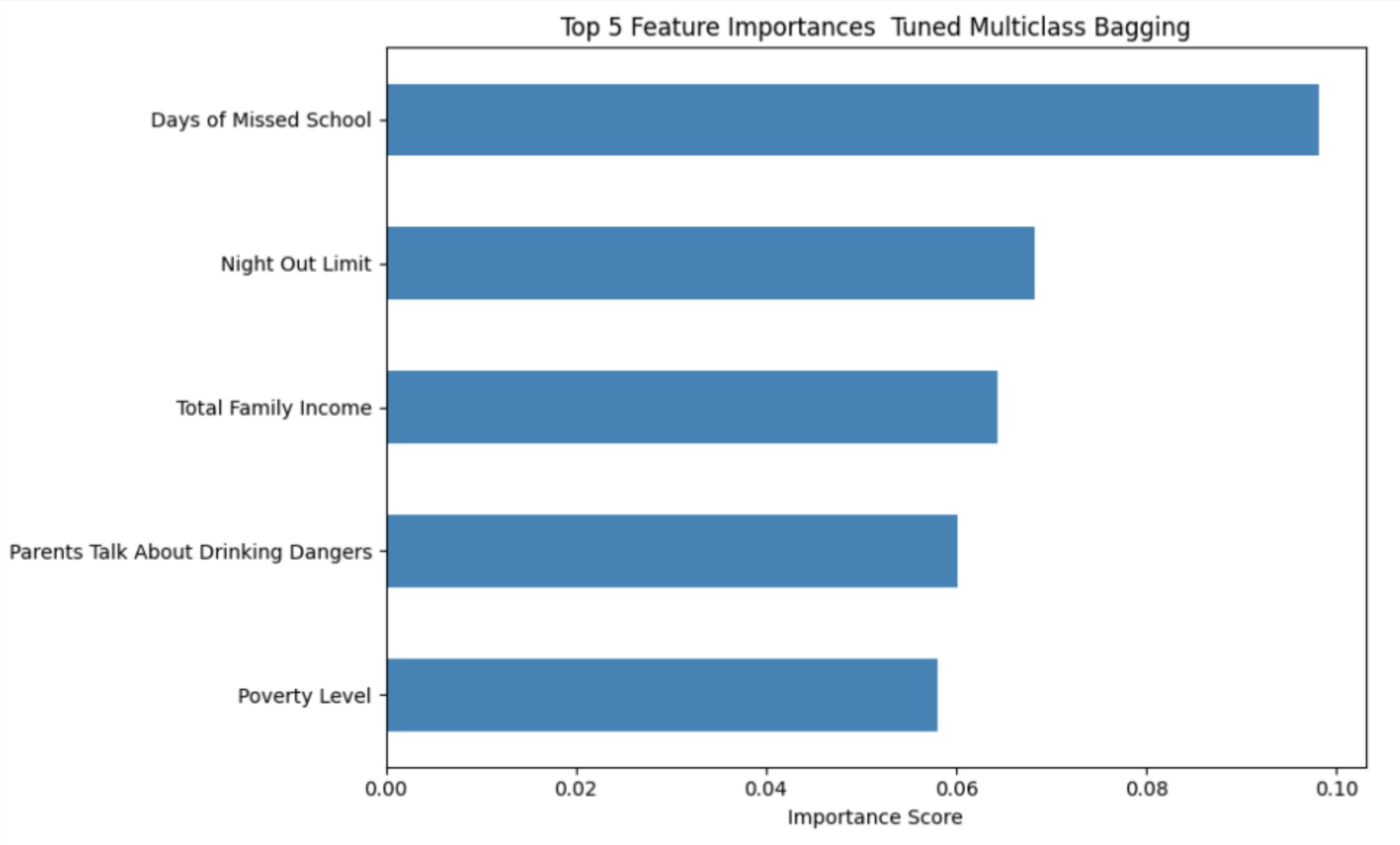
- Decision tree parameters: max_depth=10, criterion='gini', min_samples_leaf=5, min_samples_split=2
- Bagging parameters: n_estimators=15, max_samples=1.0, bootstrap=True
- Base learner struggled prior to bagging (CV Macro F1 0.31)
- Train 61.53% vs Test 32.14%
  - 29.39% gap, significant overfitting despite tuning
- Model essentially unable to identify moderate users, worst performing class (F1 0.15)

```
Train Accuracy: 0.6153
Test Accuracy: 0.3214
Difference: 0.2939
              precision    recall  f1-score   support

           0       0.27      0.25      0.26       100
           1       0.38      0.54      0.44       136
           2       0.19      0.13      0.15       103
           3       0.36      0.30      0.33       109

    accuracy                           0.32       448
   macro avg       0.30      0.30      0.30       448
weighted avg       0.30      0.32      0.31       448
```



ELBOW PLOT: OPTIMAL NUMBER OF BASE MODELS



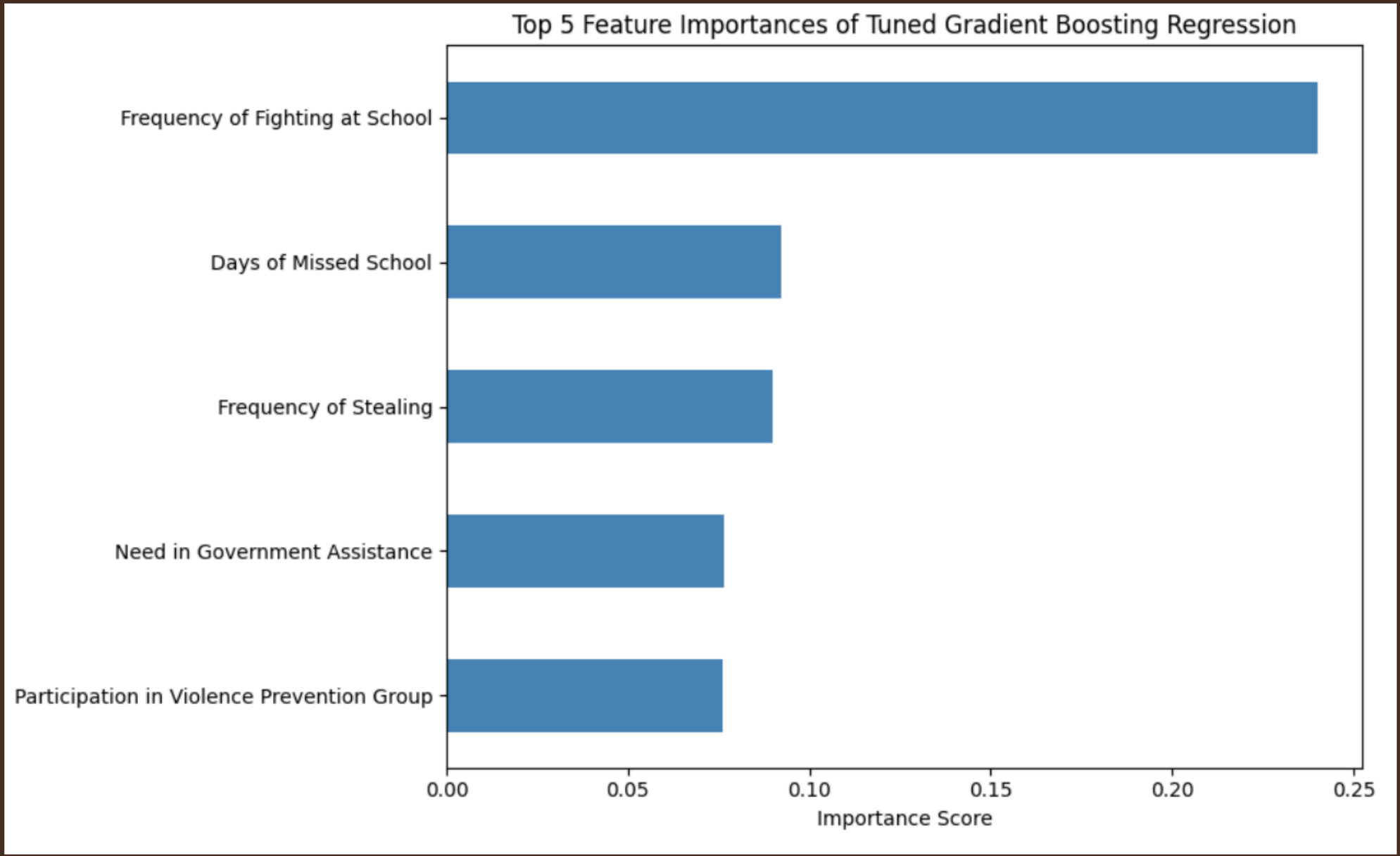

Top 5 Feature Importances Tuned Multiclass Bagging
Days of Missed School
Night Out Limit
Total Family Income
Parents Talk About Drinking Dangers
Poverty Level
0.00
0.02
0.04
0.06
0.08
0.10
Importance Score

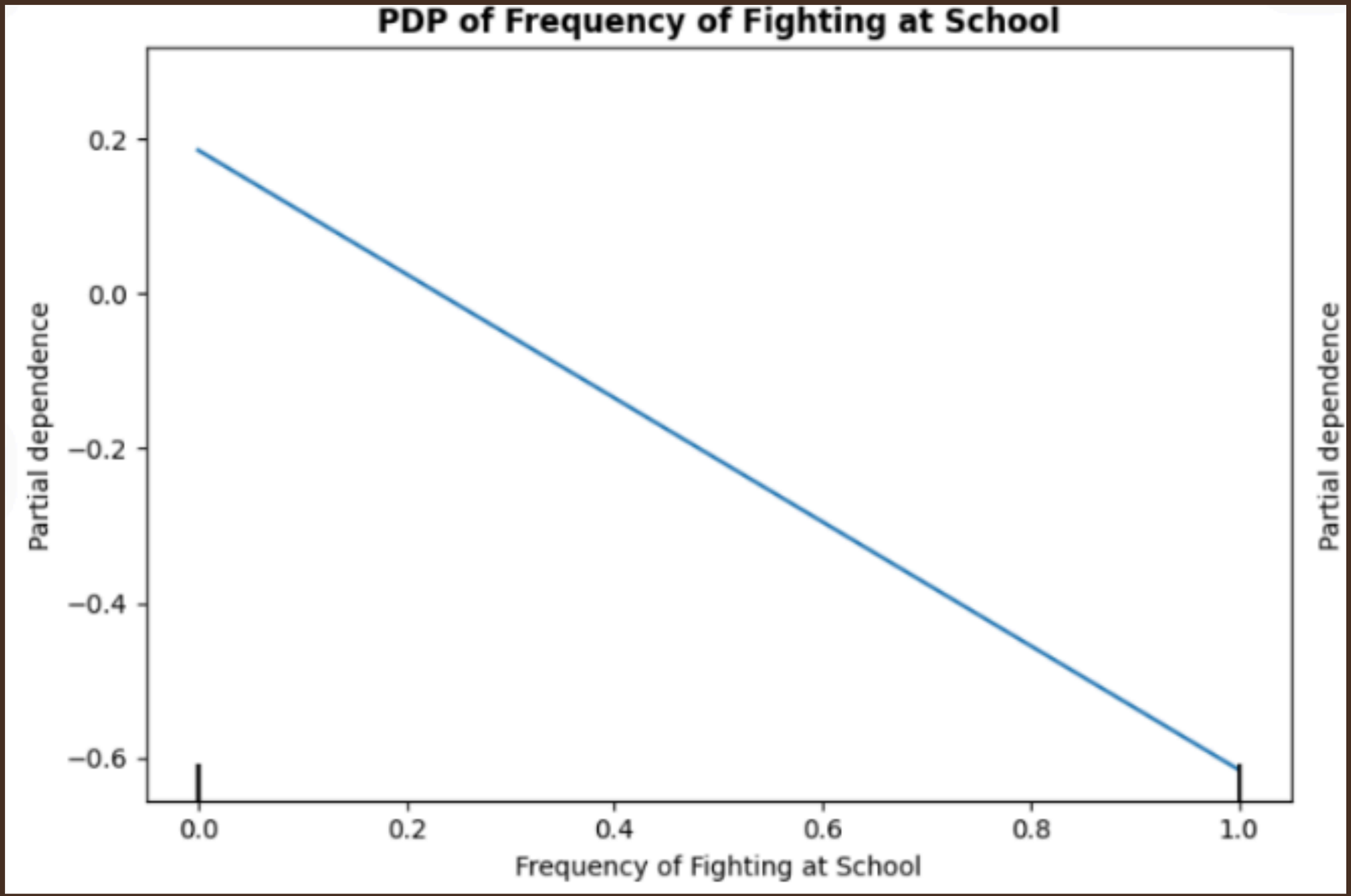
TUNED REGRESSION BOOSTING

- GridSearch conducted in structured stages: tree parameters first, then learning rate and subsample
- Best parameters: max_depth=3, min_samples_split=5, min_samples_leaf=1, learning_rate=0.1, n_estimators=30
- Shallow depth confirms simple, non-complex tree structures are optimal for this dataset
- Conservative learning_rate=0.1 with only 30 estimators
- Train RMSE: 2.322 and Test RMSE: 2.327
- Best CV RMSE: 2.4265
- CV Std: 0.0617 is very low, confirms stable and consistent generalization across all folds
- Predictions deviate from true age of first use by approximately 2.33 years on average
- Strongest and most stable model of the three with best train-test alignment and most reliable cross-validated performance

```
Train MSE:5.3924   | Train RMSE: 2.3221
Test MSE:5.4172    | Test RMSE:  2.3275
Difference:-0.0248
```

```
CV RMSE per fold: [2.4486 2.3683 2.486  2.4344 2.3128]
Mean CV RMSE: 2.4100
Std CV RMSE: 0.0617
```


Top 5 Feature Importances of Tuned Gradient Boosting Regression
Frequency of Fighting at School
Days of Missed School
Frequency of Stealing
Need in Government Assistance
Participation in Violence Prevention Group
0.00
0.05
0.10
0.15
0.20
0.25
Importance Score


PDP of Frequency of Fighting at School
0.2
0.0
−0.2
−0.4
−0.6
Partial dependence
0.0
0.2
0.4
0.6
0.8
1.0
Frequency of Fighting at School
Partial dependence

COMPARISON OF MODELS

	Binary	Multi-class	Regression
Algorithm	Decision Tree	Decision Tree + Bagging	Gradient Boosting
Target Variable	User	Frequency	Age of First Use
criterion	gini	gini	squared error
n-estimators	none	15	30
max_depth	7	10	3
min_samples_split	20	2	5
min_samples_leaf	1	5	1
learning rate	none	none	0.1
class_weight	balanced	none	none
Training Accuracy	66.64%	61.53%	none
Testing Accuracy	64.29%	32.14%	none
Macro F1	0.58	0.30	none
Weighted F1	0.68	0.31	none
Train MSE	N/A	N/A	5.392
Test MSE	N/A	N/A	5.417
Train RMSE	N/A	N/A	2.322
Test RMSE	N/A	N/A	2.327
Important feature	TV usage limit	Days of Missed School	Frequency of Fighting

LIMITATIONS

- SEVERE CLASS IMBALANCE
- MISSING DATA
- SCOPE RESTRICTION
- ARTIFICIAL CLASS BOUNDARIES
- BINARY FEATURE GRANULARITY
- ELBOW PLOT INSTABILITY
- CROSS-SECTIONAL DATA

DISCUSSION OF FACTORS

PARENT INVOLVEMENT	LIMITING TV USAGE, OFFERING PRAISE, MONITORING NIGHTTIME ACTIVITIES, DISCUSSING ALCOHOL DANGERS
SCHOOL ENGAGEMENT	DAYS OF MISSED SCHOOL
DELINQUENT BEHAVIOR	FREQUENCY OF STEALING AND FIGHTING
SOCIOECONOMIC	POVERTY LEVEL, TOTAL FAMILY INCOME, GOVERNMENT ASSISTANCE NEED

CONCLUSION

- Models applied:
  - Binary decision tree: meaningful generalization despite severe class imbalance
  - Multiclass bagging: weakest performer, limited by dataset and class definition constraints.
  - Regression gradient boosting: strongest performer
- Consistent predictors across all models:
  - Parental involvement
  - School engagement
  - Delinquent behavior
  - Socioeconomic conditions
- Youth alcohol use is as much a social phenomenon as a behavioral one. Prevention requires multi-level, community-wide approaches

CITATIONS

Google Developers. (2025, August 28). Datasets: Class-imbalanced datasets. Google Machine Learning Crash Course. Minority classes (non-alcohol users) are underrepresented.
GeeksforGeeks. (n.d.). Comprehensive guide to classification models in scikit-learn. Minority classes (non-alcohol users) are underrepresented.
Towards Data Science. (n.d.). Bagging classifier and regressor in scikit-learn. Medium. Minority classes (non-alcohol users) are underrepresented.
scikit-learn developers. (n.d.). sklearn.ensemble.RandomForestRegressor. scikit-learn. Minority classes (non-alcohol users) are underrepresented.